

Discrete Hodge Theory on Graphs

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1 Homology

1.1 Simplicial complexes and Δ -complexes

1.1.1

Definition 1.1: (non-oriented) Simplex

Let $v_0, v_1, \dots, v_p \in \mathbb{R}^n$ be $p + 1$ affine independent ^a points in $\mathbb{R}^n$. A **geometric p -simplex** is a subset of $\mathbb{R}^n$, denoted $[v_0, v_1, \dots, v_p]$,

$$[v_0, v_1, \dots, v_p] := \left\{ \sum_{i=0}^p t_i v_i : \forall i, 0 \leq t_i \leq 1, \sum_{i=0}^p t_i = 1 \right\}$$

If write $S = \{v_0, v_1, \dots, v_p\}$, then denote $\Delta_p^S := [v_0, v_1, \dots, v_p]$.
The **standard p -simplex**, denoted Δ_p , is defined as

$$\Delta_p := [e_0, e_1, \dots, e_p]$$

where e_i s are the standard basis vectors for $\mathbb{R}^p$.

^aThis can be thought as: $v_1 - v_0, \dots, v_p - v_0$ are linearly independent.

Definition 1.2: (Oriented) Simplex

Let $T = \{u_0, u_1, \dots, u_n\}$ and R be a commutative ring. An **oriented p -simplex** of X under R is an element of p -exterior product of the free module $R[T]$, $\bigwedge^p R[T]$, denoted $[u_{i_0}, u_{i_1}, u_{i_2}, \dots, u_{i_p}]^o$, or $\Delta_p^{o,T}$,

$$[u_{i_0}, u_{i_1}, u_{i_2}, \dots, u_{i_p}]^o := u_{i_0} \wedge u_{i_1} \wedge u_{i_2} \wedge \dots \wedge u_{i_p}$$

Remark When T is the standard basis of $\mathbb{R}^n$ and $R = \mathbb{R}$, then the corresponding *standard oriented p -simplex* is denoted similarly as Δ_p^o .

1.2 Chain groups

Definition 1.3: Singular chain group

The **singular chain group of X in degree p** , denoted by $S_p(X, \mathbb{Z})$, is the *free abelian group* generated by the basis {singular p -simplex in X , σ }

Remark The operation in $S_p(X, \mathbb{Z})$ is addition. The intuition of being free abelian groups is: those standard p -simplices (or singular p -simplices) are the fundamental building blocks of more complex objects. The 'addition' here corresponds to 'splicing', and we will see how to deal with the boundary nicely when splicing in the definition of boundary operators.

More explicitly,

$$S_p(X, \mathbb{Z}) = \mathbb{Z}[\text{Hom}_{\mathbf{Top}}(\Delta_p, X)]$$

The following map is to embed a standard simplex into a simplex with one-dimension higher as a face of the higher dimensional simplex. $\forall i$, the *i -th face map* in Δ_p^o is defined to be $F_{i,p} : \Delta_{p-1}^o \rightarrow \Delta_p^o$ by

$$f_j \mapsto \begin{cases} e_j & j < i \\ e_{j+1} & j \geq i \end{cases}$$

where f_i s are basis of Δ_{p-1} and e_i s are basis of Δ_p .

So, each $F_{i,p}$ is a topological embedding with image $F_{i,p}(\Delta_{p-1}) = [e_0, \dots, \hat{e}_i, \dots, e_p]$, the face opposite e_i . In other words, $F_{i,p}$ is the map for searching and anchoring Δ_{p-1} to a certain part of the boundary of Δ_p .

Definition 1.4: Boundary of singular p -simplex

The **boundary of singular p -simplex** $\sigma : \Delta_p \rightarrow X$, denoted $\partial_p \sigma$, is the map

$$\partial_p \sigma := \sum_{i=0}^p (-1)^i \sigma \circ F_{i,p}$$

In the context of exterior algebra, the whole picture can be simplified a lot:

1.2.1 Boundary operator

Since there is a notion on every singular p -simplices, and $S_p(X, \mathbb{Z})$ is generated by all singular p -simplices σ . So, the 'boundary' can be extended to the whole singular chain group $S_p(X, \mathbb{Z})$ as an operator:

Definition 1.5: Boundary operator

For X a topological space and its singular chain group in consecutive degrees p and $p-1$, the **boundary operator** between them, denoted ∂_p , is defined to be the map:

$$\partial_p : S_p(X, \mathbb{Z}) \rightarrow S_{p-1}(X, \mathbb{Z}) \quad \sigma \mapsto \partial_p \sigma := \sum_{i=0}^p (-1)^i \sigma \circ F_{i,p}$$

Remark The standard definition of the boundary operator ∂_p explicitly imposes the alternating sign $(-1)^i$. This formulation can feel somewhat unnatural because it applies an oriented algebraic operation to standard p -simplices that are fundamentally treated as mere ordered sets without intrinsic algebraic orientation.

However, this artificiality disappears if we upgrade the algebraic structure of the chain group. As we identify an oriented simplex $[v_0, v_1, \dots, v_p]$ with the wedge product of its vertices $v_0 \wedge v_1 \wedge \dots \wedge v_p$ in an exterior algebra $\bigwedge V$, the orientation is intrinsically captured by the anti-commutative property ($v_i \wedge v_j = -v_j \wedge v_i$).

In this framework, the boundary operator ∂ does not require the explicit manual insertion of $(-1)^i$. Instead, it can be elegantly and naturally defined as an **antiderivation** (specifically, an interior contraction) on the exterior algebra. The alternating signs are no longer an artificial imposition; they emerge automatically as a natural consequence of the wedge product's permutation rules when the derivation operator passes through the elements. Boundary operator has a very important property:

Theorem 1.1: Nilpotency of boundary operator

Let $\sigma \in C_p(X)$. Then $\partial_{p-1}(\partial_p(\sigma)) = 0$.

Proof:

$$\begin{aligned}
& \partial_{p-1}(\partial_p(\sigma)) \\
&= \sum_{j=0}^{p-1} (-1)^j \partial_p(\sigma) \circ F_{j,p-1} \\
&= \sum_{j=0}^{p-1} (-1)^j \left(\sum_{i=0}^p (-1)^i \sigma \circ F_{i,p} \right) \circ F_{j,p-1} \\
&= \sum_{j=0}^{p-1} \sum_{i=0}^p (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1} \\
&= \sum_{0 \leq i \leq j \leq p-1} (-1)^{i+j} \sigma \circ F_{j,p-1} \circ F_{j,p-1} + \sum_{0 \leq j < i \leq p} (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1} \\
&= \sum_{0 \leq i \leq j \leq p-1} (-1)^{i+j} \sigma \circ F_{j,p-1} \circ F_{j,p-1} + \sum_{0 \leq j < i \leq p} (-1)^{i+j} \sigma \circ F_{j,p} \circ F_{i-1,p-1} \quad j < i \leq p \Rightarrow j \leq i-1 \leq p-1 \\
&= \sum_{0 \leq i \leq j \leq p-1} (-1)^{i+j} \sigma \circ F_{j,p-1} \circ F_{j,p-1} + \sum_{0 \leq j \leq k \leq p-1} (-1)^{k+j+1} \sigma \circ F_{j,p} \circ F_{k,p-1} \quad (k := i-1) \\
&= 0
\end{aligned}$$

□

Remark In the language of exterior algebra (oriented singular p -simplex)

Corollary 1.1: Exactness of singular chain groups

The collection of all singular chain groups of X , $\{S_p(X, \mathbb{Z})\}_p$, together with the boundary operators forms an exact sequence. More explicitly, there is a long exact sequence

$$\cdots \xrightarrow{\partial_{p+2}} S_{p+1}(X, \mathbb{Z}) \xrightarrow{\partial_{p+1}} S_p(X, \mathbb{Z}) \xrightarrow{\partial_p} S_{p-1}(X, \mathbb{Z}) \xrightarrow{\partial_{p-1}} \cdots$$

Proof: The theorem 1.1 implies that $\text{im } \partial_p \subseteq \ker \partial_{p-1}$.

□

1.3

Consider $C_n(X; R) := R[\text{Hom}(\Delta_n, X)]$. Since C_n is a free R -module, it is possible to talk about the $\bigwedge^n C_n(X; R)$.

2 Cohomology

2.1 Cochains

2.1.1 Cochain group

Definition 2.1: Cochain

Let $C_n(X; R)$ be the singular n -chain group of a topological space X over R . Let S be a commutative ring such that R is a subring of S . The n -th cochain group, denoted $C^n(X; R)$, is defined as the algebraic dual of the chain group:

$$C^n(X; R) = \text{Hom}_R(C_n(X; R), S)$$

An element $\varphi \in C^n(X; R)$ is called an n -cochain.

Remark Functionally, it is a linear map that assigns an integer value to any n -chain. In the context of discrete calculus, an n -cochain serves as the exact discrete algebraic counterpart to a differential n -form.

Example (1) When $R = S = \mathbb{R}$, then $C^n(X; \mathbb{R}) = \text{Hom}_{\mathbb{R}}(C_n(X; \mathbb{R}), \mathbb{R})$, which is isomorphic to $\text{Hom}_{\mathbb{Z}}(C_n(X; \mathbb{Z}); \mathbb{R})$ by virtue of the adjoint of Hom and $\otimes$:

$$\text{Hom}_{\mathbb{Z}}(C_n(X; \mathbb{Z}); \mathbb{R}) = \text{Hom}_{\mathbb{Z}}(C_n(X; \mathbb{Z}); \text{Hom}_{\mathbb{R}}(\mathbb{R}, \mathbb{R})) \cong \text{Hom}_{\mathbb{R}}(C_n(X; \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{R}, \mathbb{R}) = \text{Hom}_{\mathbb{R}}(C_n(X; \mathbb{R}), \mathbb{R})$$

(2) When $R = S = \mathbb{C}$, the $C^n(X; \mathbb{C}) = \text{Hom}_{\mathbb{C}}(C_n(X; \mathbb{C}), \mathbb{C})$.

2.1.2 Cochain operator

The coboundary operator $\delta^n : C^n(X; R) \rightarrow C^{n+1}(X; R)$ is defined by dualizing the boundary operator ∂_{n+1} . For any n -cochain $\varphi \in C^n(X; R)$ and any $(n+1)$ -chain $c \in C_{n+1}(X; R)$, the operator δ^n is defined by its evaluation on c :

$$(\delta^n \varphi)(c) = \varphi(\partial_{n+1} c)$$

Since the boundary operator is nilpotent ($\partial \circ \partial = 0$), it naturally forces the coboundary operator to be nilpotent as well ($\delta \circ \delta = 0$). The sequence of cochain groups connected by these coboundary operators forms the cochain complex $(C^\bullet(X; R), \delta)$.

2.1.3 Cup product

Definition 2.2: Cup product

Let X be a topological space. For cochains $\varphi \in C^p(X; \mathbb{Z})$ and $\psi \in C^q(X; \mathbb{Z})$, the **cup product** $\varphi \smile \psi \in C^{p+q}(X; \mathbb{Z})$ is defined by evaluating it on a standard $(p+q)$ -simplex $\sigma = [v_0, v_1, \dots, v_{p+q}]$. The evaluation is performed by algebraically "splitting" the simplex into a front face and a back face:

$$(\varphi \smile \psi)(\sigma) := \varphi([v_0, \dots, v_p]) \cdot \psi([v_p, \dots, v_{p+q}])$$

This operation equips the cochain complex with a graded ring structure, turning the cohomology groups into the **cohomology ring** $H^\bullet(X; \mathbb{Z})$.

3 Duality

3.1 Laplacian

3.2 Integration

If we denote the evaluation of a cochain φ on a chain c using the natural pairing notation $\langle \varphi, c \rangle$ (which acts as a discrete integral), the definition rewrites as:

$$\langle \delta \varphi, c \rangle = \langle \varphi, \partial c \rangle$$

This is the pure algebraic manifestation of the topological integration identity:

$$\int_c d\omega = \int_{\partial c} \omega$$

This duality highlights a profound symmetry: the boundary operator ∂ acts geometrically by lowering the dimension of the domain (passing from a space to its boundary), whereas the coboundary operator δ acts algebraically by raising the degree of the functional (analogous to taking the exterior derivative). The evaluation bracket seamlessly bridges the geometric space and the algebraic dual space.

4 Discrete Hodge decomposition

4.1 Clique complex